Problem Statement 1 – Library Management System

Schema
BOOK
(Book_ID, Title, Author, Category, Price)
MEMBER
(Member_ID, Member_Name, City)
ISSUE
(Issue_ID, Book_ID, Member_ID, Issue_Date, Return_Date)
Questions

1. Display all books with price greater than 500.
2. Display member details from Pune city.
3. Show book title and member name for issued books.
4. Count total books in each category.
5. Display members who issued books after '2026-01-01'.
6. Display the member who issued maximum books.
7. Display category-wise average price of books having average price greater than 400.

Problem Statement 2 – Hospital Management System

Schema
DOCTOR
(Doctor_ID, Doctor_Name, Specialization, Salary)
PATIENT
(Patient_ID, Patient_Name, Age, City)
APPOINTMENT
(Appointment_ID, Doctor_ID, Patient_ID, Appointment_Date)
Questions

1. Display all doctors with salary greater than 80000.
2. Display patient details whose age is above 50.
3. Show doctor name and patient name for appointments.
4. Count total doctors specialization-wise.
5. Display appointments after '2026-03-01'.
6. Find the doctor handling maximum appointments.
7. Display specialization-wise average salary greater than 70000.

Problem Statement 3 – Online Shopping System

Schema
CUSTOMER
(Customer_ID, Customer_Name, City)
PRODUCT
(Product_ID, Product_Name, Category, Price)
ORDERS
(Order_ID, Customer_ID, Product_ID, Quantity, Order_Date)
Questions

1. Display products with price less than 1000.
2. Display customers from Mumbai.
3. Show customer name and product name for all orders.
4. Count products category-wise.
5. Display orders placed in April 2026.
6. Find the customer who placed maximum orders.
7. Display category-wise maximum product price.

Problem Statement 4 – College Management System
Schema
STUDENT
(Student_ID, Student_Name, Branch, CGPA)
FACULTY
(Faculty_ID, Faculty_Name, Department)
COURSE
(Course_ID, Course_Name, Faculty_ID)
Questions

1. Display students with CGPA greater than 8.
2. Display faculty from IT department.
3. Show course name with faculty name.
4. Count students branch-wise.
5. Display students sorted by CGPA descending.
6. Find the branch having maximum students.
7. Display average CGPA branch-wise having average CGPA above 7.

Problem Statement 5 – Banking System
Schema
CUSTOMER
(Customer_ID, Customer_Name, City)
ACCOUNT
(Account_No, Customer_ID, Account_Type, Balance)
TRANSACTION
(Transaction_ID, Account_No, Amount, Transaction_Type)
Questions

1. Display accounts with balance greater than 50000.
2. Display customers from Nashik.
3. Show customer name with account type.
4. Count accounts type-wise.
5. Display debit transactions only.
6. Find the customer having highest balance.
7. Display account type-wise average balance greater than 30000.

Problem Statement 6 – Employee Payroll System
Schema
EMPLOYEE
(Emp_ID, Emp_Name, Department, Salary)
PROJECT
(Project_ID, Project_Name, Budget)
WORKS_ON
(Emp_ID, Project_ID, Hours)
Questions

1. Display employees with salary above 60000.
2. Display projects with budget greater than 100000.
3. Show employee name with project name.
4. Count employees department-wise.
5. Display employees sorted by salary descending.
6. Find the department with highest average salary.
7. Display employees working on more than one project.

Problem Statement 7 – Railway Reservation System

Schema

TRAIN

(Train_ID, Train_Name, Source, Destination)

PASSENGER

(Passenger_ID, Passenger_Name, Age)

RESERVATION

(Reservation_ID, Train_ID, Passenger_ID, Journey_Date)

Questions

1. Display trains starting from Pune.
2. Display passengers whose age is greater than 60.
3. Show passenger name with train name.
4. Count reservations train-wise.
5. Display reservations after ‘2026-02-01’.
6. Find the train with maximum reservations.
7. Display source-wise total trains.

Problem Statement 8 – Hotel Management System

Schema

ROOM

(Room_No, Room_Type, Charges)

CUSTOMER

(Customer_ID, Customer_Name, City)

BOOKING

(Booking_ID, Room_No, Customer_ID, CheckIn_Date)

Questions

1. Display rooms with charges greater than 3000.
2. Display customers from Delhi.
3. Show customer name with room type.
4. Count rooms type-wise.
5. Display bookings in May 2026.
6. Find the most booked room type.
7. Display average room charges type-wise.

Problem Statement 9 – University Examination System

Schema

STUDENT

(Student_ID, Student_Name, Branch)

SUBJECT

(Subject_ID, Subject_Name, Credits)

RESULT

(Student_ID, Subject_ID, Marks)

Questions

1. Display students from Computer branch.
2. Display subjects with credits greater than 3.
3. Show student name with subject name.
4. Count students branch-wise.
5. Display students scoring above 75 marks.
6. Find the topper student.
7. Display average marks subject-wise greater than 70.

Problem Statement 10 – Inventory Management System
Schema
PRODUCT
(Product_ID, Product_Name, Stock, Price)
SUPPLIER
(Supplier_ID, Supplier_Name, City)
SUPPLY
(Product_ID, Supplier_ID, Quantity)
Questions

1. Display products with stock less than 20.
2. Display suppliers from Chennai.
3. Show product name with supplier name.
4. Count products supplier-wise.
5. Display products with price between 500 and 2000.
6. Find supplier supplying maximum quantity.
7. Display average product price greater than 1000.

Problem Statement 11 – Movie Ticket Booking System
Schema
MOVIE
(Movie_ID, Movie_Name, Genre, Rating)
CUSTOMER
(Customer_ID, Customer_Name, City)
BOOKING
(Booking_ID, Movie_ID, Customer_ID, Seats)
Questions

1. Display movies with rating above 8.
2. Display customers from Hyderabad.
3. Show movie name with customer name.
4. Count movies genre-wise.
5. Display bookings with seats greater than 4.
6. Find movie with maximum bookings.
7. Display average rating genre-wise.

Problem Statement 12 – Sports Club Management System
Schema
PLAYER
(Player_ID, Player_Name, Sport, Age)
COACH
(Coach_ID, Coach_Name, Experience)
TRAINING
(Player_ID, Coach_ID, Hours)
Questions

1. Display players older than 25.
2. Display coaches with experience greater than 10 years.
3. Show player name with coach name.
4. Count players sport-wise.
5. Display players sorted by age descending.
6. Find coach training maximum players.
7. Display sport-wise average age of players.

Problem Statement 13 – Airline Reservation System
Schema
FLIGHT
(Flight_ID, Flight_Name, Source, Destination)
PASSENGER
(Passenger_ID, Passenger_Name, Passport_No)
TICKET
(Ticket_ID, Flight_ID, Passenger_ID, Fare)
Questions

1. Display flights from Mumbai.
2. Display passengers whose fare is greater than 10000.
3. Show passenger name with flight name.
4. Count flights source-wise.
5. Display tickets with fare between 5000 and 15000.
6. Find flight generating maximum revenue.
7. Display average fare flight-wise.

Problem Statement 14 – E-Learning Platform
Schema
STUDENT
(Student_ID, Student_Name, Course)
INSTRUCTOR
(Instructor_ID, Instructor_Name, Expertise)
ENROLLMENT
(Student_ID, Instructor_ID, Completion_Percentage)
Questions

1. Display students enrolled in Java course.
2. Display instructors with expertise in Python.
3. Show student name with instructor name.
4. Count students course-wise.
5. Display students with completion percentage above 80.
6. Find instructor teaching maximum students.
7. Display average completion percentage instructor-wise.

Problem Statement 15 – Vehicle Service Management System
Schema
VEHICLE
(Vehicle_ID, Owner_Name, Vehicle_Type)
MECHANIC
(Mechanic_ID, Mechanic_Name, Experience)
SERVICE
(Service_ID, Vehicle_ID, Mechanic_ID, Service_Cost)
Questions

1. Display vehicles of type Car.
2. Display mechanics with experience greater than 5 years.
3. Show owner name with mechanic name.
4. Count vehicles type-wise.
5. Display services with cost greater than 5000.
6. Find mechanic handling maximum services.
7. Display average service cost mechanic-wise greater than 3000.

Problem Statement 16 – Food Delivery Application
Schema
CUSTOMER
(Customer_ID, Customer_Name, City)
RESTAURANT
(Restaurant_ID, Restaurant_Name, Rating)
ORDERS
(Order_ID, Customer_ID, Restaurant_ID, Amount)
Questions

1. Display restaurants with rating above 4.
2. Display customers from Bangalore.
3. Show customer name with restaurant name.
4. Count total orders restaurant-wise.
5. Display orders with amount greater than 1000.
6. Find restaurant receiving maximum orders.
7. Display average order amount restaurant-wise greater than 500.

Problem Statement 17 – OTT Streaming Platform
Schema
USER
(User_ID, User_Name, Subscription_Type)
SHOWS
(Show_ID, Show_Name, Genre, Rating)
WATCH_HISTORY
(User_ID, Show_ID, Watch_Time)
Questions

1. Display shows with rating greater than 8.
2. Display users having Premium subscription.
3. Show user name with watched show name.
4. Count shows genre-wise.
5. Display watch history with watch time greater than 2 hours.
6. Find most watched show.
7. Display average show rating genre-wise.

Problem Statement 18 – Smart Parking System
Schema
VEHICLE
(Vehicle_ID, Owner_Name, Vehicle_Type)
PARKING_SLOT
(Slot_ID, Floor_No, Charges)
PARKING_RECORD
(Record_ID, Vehicle_ID, Slot_ID, Hours)
Questions

1. Display parking slots with charges greater than 100.
2. Display vehicles of type Bike.
3. Show owner name with slot number.
4. Count parking slots floor-wise.
5. Display parking records with hours greater than 5.
6. Find slot used maximum times.
7. Display average parking hours slot-wise.

Problem Statement 19 – Online Examination Portal
Schema
STUDENT
(Student_ID, Student_Name, Course)
EXAM
(Exam_ID, Subject_Name, Total_Marks)
RESULT
(Student_ID, Exam_ID, Marks_Obtained)
Questions

1. Display exams with total marks greater than 50.
2. Display students enrolled in AI course.
3. Show student name with subject name.
4. Count students course-wise.
5. Display students scoring above 80.
6. Find student obtaining highest marks.
7. Display subject-wise average marks greater than 70.

Problem Statement 20 – EV Charging Station System
Schema
VEHICLE
(Vehicle_ID, Owner_Name, Battery_Capacity)
STATION
(Station_ID, Station_Name, Location)
CHARGING_SESSION
(Session_ID, Vehicle_ID, Station_ID, Units_Consumed)
Questions

1. Display vehicles with battery capacity greater than 50.
2. Display charging stations located in Pune.
3. Show owner name with station name.
4. Count charging sessions station-wise.
5. Display sessions with units consumed greater than 30.
6. Find station with highest total units consumed.
7. Display average units consumed station-wise.

Problem Statement 21 – Freelancing Marketplace
Schema
FREELANCER
(Freelancer_ID, Freelancer_Name, Skill, Rating)
CLIENT
(Client_ID, Client_Name, Country)
PROJECT
(Project_ID, Freelancer_ID, Client_ID, Budget)
Questions

1. Display freelancers with rating above 4.5.
2. Display clients from USA.
3. Show freelancer name with client name.
4. Count freelancers skill-wise.
5. Display projects with budget greater than 50000.
6. Find freelancer handling maximum projects.
7. Display skill-wise average freelancer rating.

Problem Statement 22 – Digital Wallet System
Schema
USER
(User_ID, User_Name, City)
WALLET
(Wallet_ID, User_ID, Balance)
TRANSACTION
(Transaction_ID, Wallet_ID, Amount, Transaction_Type)
Questions

1. Display wallets with balance greater than 10000.
2. Display users from Mumbai.
3. Show user name with wallet balance.
4. Count transactions type-wise.
5. Display credit transactions only.
6. Find wallet with highest balance.
7. Display average transaction amount type-wise.

Problem Statement 23 – Telemedicine System
Schema
DOCTOR
(Doctor_ID, Doctor_Name, Specialization)
PATIENT
(Patient_ID, Patient_Name, City)
CONSULTATION
(Consultation_ID, Doctor_ID, Patient_ID, Fees)
Questions

1. Display doctors specialized in Cardiology.
2. Display patients from Pune.
3. Show doctor name with patient name.
4. Count consultations doctor-wise.
5. Display consultations with fees greater than 1500.
6. Find doctor handling maximum consultations.
7. Display specialization-wise average consultation fees.

Problem Statement 24 – Ride Sharing Application
Schema
DRIVER
(Driver_ID, Driver_Name, Rating)
RIDER
(Rider_ID, Rider_Name, City)
RIDE
(Ride_ID, Driver_ID, Rider_ID, Fare)
Questions

1. Display drivers with rating above 4.
2. Display riders from Chennai.
3. Show driver name with rider name.
4. Count rides driver-wise.
5. Display rides with fare greater than 500.
6. Find driver earning maximum fare.
7. Display average fare driver-wise.

Problem Statement 25 – Smart Gym Management System
Schema
MEMBER
(Member_ID, Member_Name, Plan_Type)
TRAINER
(Trainer_ID, Trainer_Name, Experience)
SESSION
(Session_ID, Member_ID, Trainer_ID, Calories_Burned)
Questions

1. Display members with Premium plan.
2. Display trainers with experience greater than 3 years.
3. Show member name with trainer name.
4. Count members plan-wise.
5. Display sessions with calories burned greater than 500.
6. Find trainer handling maximum sessions.
7. Display average calories burned trainer-wise.

MongoDB Problem Statement 1 – Library Management System
Collection: books

```
{
  "book_id": 101,
  "title": "Database System",
  "author": "Korth",
  "category": "Education",
  "price": 650,
  "copies": 5
}
```

Queries

1. Display all books.
2. Display books with price greater than 500.
3. Display books belonging to Education category.
4. Display only title and author fields.
5. Count total books.
6. Update copies of a specific book.
7. Delete books with copies less than 2.

MongoDB Problem Statement 2 – Hospital Management System

Collection: patients

```
{
  "patient_id": 201,
  "patient_name": "Rahul Sharma",
  "age": 45,
  "disease": "Diabetes",
  "city": "Pune"
}
```

Queries

1. Display all patients.
2. Display patients with age greater than 40.
3. Display patients from Pune.
4. Display only patient name and disease.
5. Count total patients.
6. Update disease of a patient.
7. Delete patients with age less than 18.

MongoDB Problem Statement 3 – Online Shopping System

Collection: products

```
{
  "product_id": 301,
  "product_name": "Laptop",
  "category": "Electronics",
  "price": 55000,
  "stock": 12
}
```

Queries

1. Display all products.
2. Display products with price greater than 50000.
3. Display products in Electronics category.
4. Display only product name and price.
5. Count total products.
6. Update stock of a product.
7. Delete products with stock equal to 0.

MongoDB Problem Statement 4 – College Management System

Collection: students{

```
  "student_id": 401,
  "student_name": "Amit Patil",
  "branch": "IT",
  "cgpa": 8.5,
  "city": "Mumbai"
}
```

Queries

1. Display all students.
2. Display students with CGPA greater than 8.
3. Display students from Mumbai.
4. Display only student name and branch.
5. Count total students.
6. Update CGPA of a student.
7. Delete students with CGPA less than 5.

MongoDB Problem Statement 5 – Banking System

Collection: accounts

```
{
 "account_no": 5001,
 "customer_name": "Neha Joshi",
 "account_type": "Savings",
 "balance": 85000,
 "city": "Nashik"
}
```

Queries

1. Display all accounts.
2. Display accounts with balance greater than 50000.
3. Display customers from Nashik.
4. Display only customer name and balance.
5. Count total accounts.
6. Update balance of an account.
7. Delete accounts with balance less than 1000.

MongoDB Problem Statement 6 – Food Delivery Application

Collection: orders

```
{
 "order_id": 601,
 "customer_name": "Rohan",
 "restaurant_name": "Spice Hub",
 "amount": 850,
 "status": "Delivered"
}
```

Queries

1. Display all orders.
2. Display orders with amount greater than 500.
3. Display delivered orders.
4. Display only customer name and amount.
5. Count total orders.
6. Update order status.
7. Delete cancelled orders.

MongoDB Problem Statement 7 – OTT Streaming Platform

Collection: shows{

```
 "show_id": 701,
 "show_name": "Dark Code",
 "genre": "Thriller",
 "rating": 8.9,
 "language": "English"
}
```

Queries

1. Display all shows.
2. Display shows with rating greater than 8.
3. Display Thriller genre shows.
4. Display only show name and rating.
5. Count total shows.
6. Update language of a show.
7. Delete shows with rating less than 5.

MongoDB Problem Statement 8 – Smart Parking System

Collection: parking

```
{
 "vehicle_no": "MH14AB1234",
 "owner_name": "Kiran",
 "slot_no": 12,
 "hours": 4,
 "charges": 200
}
```

Queries

1. Display all parking records.
2. Display records with charges greater than 100.
3. Display records with parking hours greater than 3.
4. Display only owner name and slot number.
5. Count total parking records.
6. Update parking charges.
7. Delete parking records with hours equal to 0.

MongoDB Problem Statement 9 – Ride Sharing Application

Collection: rides

```
{
 "ride_id": 901,
 "driver_name": "Suresh",
 "rider_name": "Anjali",
 "fare": 450,
 "city": "Pune"
}
```

Queries

1. Display all rides.
2. Display rides with fare greater than 300.
3. Display rides from Pune.
4. Display only driver name and fare.
5. Count total rides.
6. Update ride fare.
7. Delete rides with fare less than 50.

MongoDB Problem Statement 10 – Smart Gym Management System

Collection: members{

```
 "member_id": 1001,
 "member_name": "Rahul",
 "plan_type": "Premium",
 "trainer": "Akash",
 "calories_burned": 650
}
```

Queries

1. Display all gym members.
2. Display members with calories burned greater than 500.
3. Display Premium plan members.
4. Display only member name and trainer.
5. Count total members.
6. Update trainer name.
7. Delete members with calories burned less than 100.